## Notes regarding the exam in Bioinformatics Algorithms:

Duration: about 90 minutes
Format: written only

The exam will contain both

- **practical exercises** (e.g., sequence alignment, Burrows-Wheeler transformation, construction of graph s like suffix trees, etc.)
- **theoretical questions** (e.g., brief description of a specific algorithmic approach, complexity, some *very basic* notions about the underlying molecular biology, the most *basic* equations, etc.)

Practical exercises:

For the practical part, many websites are available that allow you to arbitrarily invent some exercises and verify the correctness of your obtained result:

a) Burrows-Wheeler transform (encoding and decoding):
https://calcoolator.eu/burrows-wheeler-transform-encoder-decoder-

b) Needlman-Wunsch (global sequence alignment):
http://rna.informatik.uni-freiburg.de/Teaching/index.jsp?toolName=Needleman-Wunsch

c) Smith-Waterman (local sequence alignment):
http://rna.informatik.uni-freiburg.de/Teaching/index.jsp?toolName=Smith-Waterman

d) Creation of a suffix tree:
https://visualgo.net/en/suffixtree
(Note that this tool accepts only uppercase letters plus the $ as termination symbol)

e) Trie:
https://gallery.selfboot.cn/en/algorithms/trie
(You have to insert single words, i.e., to build a *suffix* trie you need to insert all single suffixes here!)

Theoretical questions:

Theoretical questions are mostly aimed at verifying your understanding of a topic rather than the practical capacity to execute an algorithm or perform a procedure.

The following are just a few *examples* of possiblequestions that might asked in the exam (as such or similar, no guarantee of completeness of this list!):

a) When performing a local sequence alignment, how can you identify multiple equivalent solutions in the alignment matrix after you filled it with dynamic programming?

b) What fourth option/choice do you have for each cell when performing a local sequence alignment instead of a global sequence alignment, and what does that fourth/choice option mean?

c) A suffix tree should be as compact as possible. What are the major features that reduce the size complexity?

d) How is the starting sequence of the “center star” algorithm chosen?

e) Briefly describe the “central dogma” of molecular biology.

f) In the context of a Hidden Markov Model, what is the major difference between the Viterbi decoding and the posterior decoding? (in very brief words)